

Day 8 — TNPSC Group 2A Study Material

Part A – General Studies: Environment, Ecology & Latest Inventions in Science and Technology

READING MATERIAL

The final General Science clause, covering environment/ecology plus 'latest inventions' (which in practice means established science-current-affairs facts, not literally cutting-edge news — those go stale fast and should be refreshed close to the actual exam date rather than memorized far in advance).

Q: Real exam: What is Tamil Nadu's major renewable energy source?

A: Windmill (wind energy) — not coal, oil and gas, or nuclear power. Tamil Nadu has historically been a leader in wind power generation in India.

Q: Real exam: What are the three main aspects of sustainable development?

A: Social, Environment, and Economy (the 'triple bottom line') — not Natural Resources Management, Pollution Control, or Technology specifically; those are tools/concerns within the three main pillars, not the pillars themselves.

Q: What is an ecosystem, and what are its two main components?

A: An ecosystem is a community of living organisms interacting with their physical environment. Its two main components are biotic (living: plants, animals, microbes) and abiotic (non-living: soil, water, air, sunlight).

Q: What is biodiversity, and why does it matter?

A: The variety of life (species, genetic variation, ecosystems) in a given area. High biodiversity makes ecosystems more resilient/stable and supports more ecological services (pollination, soil fertility, climate regulation).

Q: What are the main types of environmental pollution?

A: Air pollution, water pollution, soil pollution, noise pollution, and thermal pollution — each with its own sources (vehicle/industrial emissions, sewage/industrial discharge, pesticides/plastics, loud machinery/traffic, heated industrial discharge into water bodies) and control approaches.

Q: What causes the greenhouse effect, and which gases are the main contributors?

A: Certain atmospheric gases trap heat radiated from the Earth's surface, warming the planet. Main greenhouse gases: carbon dioxide (CO₂), methane (CH₄), nitrous oxide, and water vapour. Excess greenhouse gases from human activity (burning fossil fuels, deforestation) drive climate change.

Q: What is the difference between in-situ and ex-situ biodiversity conservation?

A: In-situ conservation protects species within their natural habitat (e.g., national parks, wildlife sanctuaries, biosphere reserves). Ex-situ conservation protects species outside their natural habitat (e.g., zoos, seed banks, botanical gardens).

MOCK TEST (WITH ANSWERS)

1. The major renewable energy source in Tamil Nadu is:

- A. Windmill ✓**
- B. Coal
- C. Oil and Gas
- D. Nuclear power

Explanation: Windmill (wind energy). [easy, question-bank-2025]

2. Sustainable development is said to have three main aspects, namely:

- A. Social, Environment, Natural Resources Management
- B. Social, Environment, Economy ✓**
- C. Social, Environment, Pollution Control
- D. Social, Environment, Technology

Explanation: Social, Environment, Economy — the 'triple bottom line'. [medium, question-bank-2026]

3. An ecosystem's living components (plants, animals, microbes) are called:

- A. Abiotic components

B. Biotic components ✓

C. Trophic components

D. Climatic components

Explanation: Biotic components (as opposed to abiotic, non-living components). [easy, authored]

4. Which of the following is NOT typically classified as a major type of environmental pollution?

A. Air pollution

B. Noise pollution

C. Thermal pollution

D. Gravitational pollution ✓

Explanation: 'Gravitational pollution' isn't a recognized category — air, water, soil, noise, and thermal pollution are the standard types. [easy, authored]

5. Which of the following is the MAIN greenhouse gas contributing to global warming from human activity?

A. Nitrogen

B. Oxygen

C. Carbon dioxide ✓

D. Argon

Explanation: Carbon dioxide (CO₂), mainly from burning fossil fuels. [easy, authored]

6. Protecting an endangered species within a wildlife sanctuary (its natural habitat) is an example of:

A. Ex-situ conservation

B. In-situ conservation ✓

C. Captive breeding only

D. Genetic engineering

Explanation: In-situ conservation — protection within the natural habitat. [medium, authored]

7. Storing seeds in a seed bank, away from their natural growing environment, is an example of:

A. In-situ conservation

B. Ex-situ conservation ✓

C. Bioremediation

D. Afforestation

Explanation: Ex-situ conservation — protection outside the natural habitat. [medium, authored]

Part B – Aptitude & Mental Ability: Area

READING MATERIAL

Area is 1 of 10 named Aptitude sub-skills. Real exam evidence shows a strong preference for applied/real-world framing (fencing cost, carpet cost, crossroads, a grazing horse) over bare formula plug-ins — and frequently combines area with a real-world cost-per-unit calculation.

Q: Method — area of a triangle from coordinates: Find the area of the triangle with vertices (0,0), (2,0), (0,2).

A: When two sides lie along the axes (a right triangle at the origin), area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2 \times 2 = 2$ sq. units. (The general coordinate formula $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ also works, but recognizing the right-angle-at-origin shortcut is faster.)

Q: Real exam: A rectangular ground 80m $\times$ 60m has two equal-width cross roads (one parallel to each side). Their combined area is 675 sq.m. Find the road width.

A: If width= w , the two roads' combined area = $80w + 60w - w^2$ (subtracting the overlapping square where they cross, counted twice otherwise) = $140w - w^2$. Set equal to 675: $w^2 - 140w + 675 = 0$. Solving gives $w=5$ (the other root, 135, is too large to be physically valid for an 80 $\times$ 60 ground).

Q: Real exam: Find the perimeter of a semicircle of radius 28cm.

A: Perimeter of a semicircle = $\pi r + 2r$ (the curved half plus the straight diameter edge). Using $\pi \approx 22/7$: $(22/7 \times 28) + (2 \times 28) = 88 + 56 = 144$ cm. (Common mistake: forgetting to add the straight diameter edge, which would wrongly give just 88cm.)

Q: Real exam: A horse is tethered by a 14m rope to one corner of a 60m $\times$ 42m rectangular field. How much area is left UNGRAZED?

A: The horse can graze a quarter-circle of radius 14m (since it's tied at a corner, with two field-edges as boundaries): grazed area = $\frac{1}{4} \times \pi \times 14^2 = \frac{1}{4} \times (22/7) \times 196 = 154$ sq.m. Total field area = $60 \times 42 = 2520$ sq.m. Ungrazed = $2520 - 154 = 2366$ sq.m.

Q: Real exam: Area of a rhombus with diagonals 6cm and 8cm?

A: Area of rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 6 \times 8 = 24$ cm².

Q: Real exam: If fencing a circular park at ₹5/metre costs ₹1,100 total, find the park's radius.

A: Total fencing length (= circumference) = $₹1,100 \div ₹5/\text{m} = 220$ m. Circumference = $2\pi r \rightarrow 220 = 2 \times (22/7) \times r \rightarrow r = 220 \times 7/44 = 35$ m.

Q: Real exam: Find the cost of carpeting a room 13m long, 9m wide, using a carpet 75cm wide, at Rs.8 per metre (of carpet length).

A: Room area = $13 \times 9 = 117$ m². Carpet width = 0.75m, so length of carpet needed = Room area $\div$ carpet width = $117/0.75 = 156$ m. Cost = $156 \times \text{Rs.}8 = \text{Rs.}1,248$.

Q: Real exam (related, surface area of a 3D shape): The curved surface area of a right circular cylinder of height 7cm is 66 cm². Find its diameter.

A: CSA = $2\pi rh \rightarrow 66 = 2 \times (22/7) \times r \times 7 = 44r \rightarrow r = 1.5$ cm $\rightarrow$ diameter = 3cm.

MOCK TEST (WITH ANSWERS)

1. Find the area of the triangle with vertices (0,0), (5,0), (0,4).

A. 10 sq. units ✓

B. 20 sq. units

C. 9 sq. units

D. 18 sq. units

Explanation: Right triangle at origin: area = $\frac{1}{2} \times 5 \times 4 = 10$ sq. units. [easy, authored]

2. Find the perimeter of a semicircle of radius 21cm (use $\pi \approx 22/7$).

A. 66 cm

B. 108 cm ✓

C. 132 cm

D. 96 cm

Explanation: Perimeter = $\pi r + 2r = (22/7 \times 21) + (2 \times 21) = 66 + 42 = 108$ cm. [medium, authored]

3. Find the area of a rhombus with diagonals 10cm and 12cm.

A. 60 cm² ✓

B. 120 cm²

C. 30 cm²

D. 100 cm^2

Explanation: Area = $\frac{1}{2} \times 10 \times 12 = 60\text{ cm}^2$. [easy, authored]

4. Fencing a circular garden at ₹4 per metre costs ₹880 in total. Find the garden's radius (use $\pi \approx 22/7$).

A. 28 m

B. 35 m ✓

C. 22 m

D. 14 m

Explanation: Circumference = $880/4 = 220\text{m}$. $r = 220 \times 7 / (2 \times 22) = 1540/44 = 35\text{m}$. [medium, authored]

5. Find the cost of carpeting a room 10m long and 8m wide with a carpet 80cm wide, at Rs.10 per metre.

A. Rs.800

B. Rs.1,000 ✓

C. Rs.1,200

D. Rs.960

Explanation: Room area = 80 m^2 . Carpet length needed = $80/0.8 = 100\text{m}$. Cost = $100 \times \text{Rs.}10 = \text{Rs.}1,000$. [medium, authored]

6. A horse is tethered by a 7m rope to a corner of a $20\text{m} \times 15\text{m}$ rectangular field. Find the grazed area (use $\pi \approx 22/7$).

A. 38.5 m^2 ✓

B. 44 m^2

C. 49 m^2

D. 77 m^2

Explanation: Quarter circle of radius 7m: $\frac{1}{4} \times (22/7) \times 49 = 38.5\text{ m}^2$. [medium, authored]

Part C – General English: Poem: "The Listeners" — Walter de la Mare

READING MATERIAL

A famously eerie, mysterious poem about a Traveller who knocks at a lone house's moonlit door, calls out, and gets no answer — except from unseen 'phantom listeners'. Real coaching-site evidence tests plot, atmosphere/tone, and figure-of-speech on specific lines.

Q: What happens in "The Listeners"?

A: A Traveller knocks on the door of a lone house at night (moonlit setting) and calls out 'Is there anybody there?' No human answers, but the house is full of unseen 'phantom listeners' who hear him. Perplexed that no one answers, the Traveller eventually leaves.

Q: What is the overall tone and setting of the poem?

A: Mysterious and eerie tone, set in a moonlit, silent, forest-adjacent house at night — atmosphere is the poem's central achievement.

Q: What does the phrase "phantom listeners" use as a figure of speech?

A: Personification — listeners that are 'phantom' (not physically present/human) are nonetheless given the human act of listening.

Q: What does "His horse in the silence champed the grasses" suggest, atmospherically?

A: That the silence is so deep that even a small, ordinary sound (a horse chewing grass) becomes noticeable — emphasizing just how quiet and still the setting is.

Q: Identify the figure of speech in "leaf-fringed sill."

A: Alliteration — repeated 'f' sound in 'leaf-fringed'.

Q: Why is the Traveller described as "perplexed"?

A: Because no one answers his call, despite the house seeming to be 'full' of listeners (the phantom listeners hear him, but don't respond).

Q: What does "Is there anybody there?" symbolize/express?

A: A desire for connection/contact — the Traveller's call represents a basic human wish to be heard and answered.

Q: What does the 'moonlit door' contribute to the poem?

A: Mystery and atmosphere — the moonlight visually sets the eerie, dreamlike, uncertain mood of the whole scene.

MOCK TEST (WITH ANSWERS)

1. "The Listeners" was written by:

- A. William Wordsworth
- B. Robert Frost
- C. Walter de la Mare ✓**
- D. Sarojini Naidu

Explanation: Walter de la Mare. [easy, web-research]

2. What is the Traveller doing at the start of the poem?

- A. Escaping a war
- B. Searching for food
- C. Knocking on a door ✓**
- D. Singing in the woods

Explanation: Knocking on the door of a lone house. [easy, web-research]

3. Who actually hears the Traveller's call, per the poem?

- A. The villagers
- B. His horse
- C. Phantom listeners ✓**
- D. Birds

Explanation: Phantom listeners — unseen presences inside the house. [medium, web-research]

4. "Phantom listeners" is an example of which figure of speech?

- A. Metaphor
- B. Simile

C. Personification ✓

D. Alliteration

Explanation: Personification — non-human/unseen presences given the human act of listening. [medium, web-research]

5. "His horse in the silence champed the grasses" suggests:

A. The horse is wild

B. Nature is loud

C. The silence is so deep that small sounds are noticeable ✓

D. The horse is tired

Explanation: Emphasizes the depth of the surrounding silence. [medium, web-research]

6. "Leaf-fringed sill" is an example of:

A. Simile

B. Alliteration ✓

C. Irony

D. Metaphor

Explanation: Alliteration — repeated 'f' sound. [medium, web-research]

7. The overall tone of "The Listeners" is best described as:

A. Joyful

B. Mysterious and eerie ✓

C. Romantic

D. Humorous

Explanation: Mysterious and eerie. [easy, web-research]

8. Why is the Traveller "perplexed" in the poem?

A. He sees a ghost directly

B. No one answers his call ✓

C. His horse runs away

D. He's lost his way

Explanation: No one answers, despite the house seemingly being occupied. [medium, web-research]